

Veera Jeeshitha Kolla

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EDUCATION

Rutgers University, New Brunswick, NJ

Master of Science in Data Science, 3.8/4.0

May, 2026

Bangalore Institute of Technology, Bangalore, India

Bachelor of Engineering, Computer Science, 8.8/10.0

May, 2024

TECHNICAL SKILLS

Languages & Libraries: Python, TypeScript, JavaScript, R, C, SQL, Bash, spaCy, scikit-learn, CatBoost, HuggingFace Transformers, pandas, Matplotlib, Java

Frameworks & Tools: FastAPI, Node.js, Express, React, React Native (Expo), Docker, SQLAlchemy, Zod, Jest, Swagger UI, Power BI, Tableau, WordPress, Apache Kafka, Kubernetes

Databases: PostgreSQL, MongoDB Atlas, SQLite

Concepts: NLP, Machine Learning, Agentic AI, LLM Tool Use, Anomaly Detection, Federated Learning, SMOTE, Feature Engineering, ETL Pipelines, REST APIs, Full-Stack Development, Data Visualization, Model Deployment

AI-Assisted Development Tools: GitHub Copilot, Claude Code, ChatGPT, Cursor AI

PROFESSIONAL EXPERIENCE

Enterprise Building Training Solutions

AI & Data Intern

Oct 2023 - Jun 2024

Tools & Technologies: Python, scikit-learn, pandas, NumPy, Matplotlib, Jupyter Notebook, SQL, Excel, Git

- Built and evaluated **machine learning models** in Python (scikit-learn, pandas) for classification and prediction tasks, benchmarking multiple algorithms to select the best-performing approach
- Designed **data pipelines** for **cleaning, preprocessing, and feature engineering** on real-world business datasets, significantly reducing manual data preparation effort
- Developed **solutions to automate internal workflows**, working with stakeholders to translate business requirements into technical deliverables
- Presented **model results and data insights** to non-technical stakeholders through **visualizations and reports**, informing business decisions

Sentari AI

SDE Intern

July 2025 - Aug 2025

Tools & Technologies: Python, React Native (Expo), TypeScript, React, spaCy, WordPress, REST APIs, GitHub

- Built **NLP-based** contradiction and thought detection using **spaCy**.
- Developed and maintained **WordPress website** with custom pages and scheduled blog posts.
- Implemented delete account feature in **iOS app** using **React Native (Expo)**.
- Created **Contradiction & Conflict Insight Card** with **Python backend** and **React/TypeScript frontend**.
- Pushed production code to **GitHub**, raised and iterated on pull requests, and incorporated code review feedback.

RESEARCH PROJECTS

Job Match Assistant

2026

Tech Stack: Node.js, Express, TypeScript, React, Anthropic Claude API (tool use), Zod, Jest, Vite

- Built a full-stack web application that analyzes job descriptions against a structured candidate profile using an **agentic Claude pipeline**, returning a match score, gap analysis, and tailored resume bullet suggestions.
- Engineered a multi-step **agent loop on Anthropic's tool-use API** orchestrating three tools: LLM-based JD parsing, a deterministic keyword-scoring function with normalization and alias matching, and profile-grounded bullet drafting with a hard iteration cap to prevent runaway loops.
- Developed a **Node.js/Express REST API** in strict **TypeScript** with **Zod** request validation and centralized error handling that prevents API keys and raw errors from reaching the client.
- Wrote offline unit and integration tests with **Jest** and **Supertest**, fully mocking the Anthropic SDK so the suite runs deterministically without consuming API credits.

Real-Time Sentiment Analysis Dashboard

2026

Tech Stack: Apache Kafka, FastAPI, React, PostgreSQL, HuggingFace Transformers (DistilBERT), Docker

- Built an end-to-end real-time sentiment analysis pipeline using **Apache Kafka** to ingest and process streaming text data.
- Implemented a **DistilBERT-based NLP sentiment classifier** with HuggingFace Transformers and integrated it into a Kafka consumer service.
- Developed RESTful APIs with FastAPI to deliver live sentiment metrics and historical insights stored in **PostgreSQL**.
- Designed and deployed an interactive React dashboard with real-time visualizations, fully containerized using **Docker Compose**.

FEVER Fact Verification using NLP Models

2025

Tech Stack: Python, HuggingFace Transformers, BERT, DistilBERT, RoBERTa, scikit-learn, TF-IDF, PyTorch

- Implemented an end-to-end fact verification pipeline for the FEVER dataset to classify claims as SUPPORTS, REFUTES, or NOT ENOUGH INFO using Wikipedia evidence.
- Built and evaluated baseline ML models (TF-IDF + Logistic Regression, SVM, Random Forest), achieving up to **71.9%** accuracy with SVM.
- Fine-tuned transformer models (BERT, DistilBERT, RoBERTa), improving performance to **approximately 87% accuracy and macro-F1**, significantly outperforming baselines.
- Conducted detailed error analysis using confusion matrices and class-wise error distributions to compare model behavior and identify challenging claim categories.

Global COVID-19 Dashboard

2025

Tech Stack: SQLAlchemy, PostgreSQL, pandas, Power BI, REST APIs

- Built a real-time **ETL pipeline** to extract COVID-19 data via public APIs, transform it using **pandas**, and load it into **PostgreSQL** using SQLAlchemy for centralized storage.
- Developed a visually interactive **Power BI dashboard** featuring map visuals, time-series graphs, and filters to track and compare case data by country and continent.
- Leveraged **SQL scripts** to enable daily data refresh and dynamic views, ensuring the dashboard consistently reflected live, accurate data.

Insurance Claim Prediction

2023

Tech Stack: Python, scikit-learn, pandas, SMOTE, Matplotlib

- Cleaned and preprocessed a dataset of 1,338 samples, reducing data noise by 17% via **null handling and feature elimination** in pandas.
- Tackled class imbalance by applying **SMOTE**, increasing the dataset by 5.5% and improving model generalization.
- Trained and evaluated six machine learning models including **Logistic Regression, Random Forest, and Gradient Boosting**, with Gradient Boosting achieving the best performance based on F1-score.
- Visualized and interpreted results using **Matplotlib**, reinforcing understanding of model behavior and production readiness

Network Intrusion Detection Using Federated Learning

2024

Tech Stack: Python, CatBoost, Bagging Classifier, SMOTE, scikit-learn

- Collaborated in a 4-member team to build a **decentralized machine learning system** for detecting network intrusions, implementing a federated learning setup using **CatBoost** (local model) and a **Bagging Classifier** (global model).
- Handled large-scale intrusion datasets through **NULL handling, outlier removal, and feature engineering**, enhancing data quality and model reliability.
- Addressed class imbalance using **SMOTE**, expanding the minority class representation to 1,500 samples per category.
- Achieved **98% test accuracy** and published the results in the *Compliance Engineering Journal*, demonstrating expertise in cybersecurity-focused AI.

HONORS AND ACTIVITIES

Advanced SQL

Network Intrusion Detection using Federated Learning

*Certificate of completion from Great Learning
Publication by Compliance Engineering Journal*